Rajalakshmi Engineering College

FUNDAMENTALS OF DATA SCIENCE LAB

EXPERIMENT 17

Understand linear regression for a given dataset

Experiment 17: Understand linear regression for a given dataset

Aim

To apply linear regression to the given dataset.

Source Code

```
import numpy as np
import pandas as pd
df=pd.read_csv('Salary_data.csv')
df
df.info()
df.dropna(inplace=True)
df.info()
df.describe()
features=df.iloc[:,[0]].values
label=df.iloc[:,[1]].values
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_st
from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
model.score(x_train,y_train)
model.score(x_test,y_test)
model.coef_
model.intercept_
import pickle
pickle.dump(model,open('SalaryPred.model','wb'))
model=pickle.load(open('SalaryPred.model','rb'))
yr_of_exp=float(input("Enter Years of Experience: "))
yr_of_exp_NP=np.array([[yr_of_exp]])
Salary=model.predict(yr_of_exp_NP)
print("Estimated Salary for {} years of experience is {}: " .format(yr_of_exp,Salary)
```

Output - Experiment 17

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
YearsExperience Salary
count 30.000000 30.000000
mean 5.313333 76003.000000
std 2.837888 27414.429785
min 1.100000 37731.000000
25% 3.200000 56720.750000
50% 4.700000 65237.000000
75% 7.700000 100544.750000
max 10.500000 122391.000000
□LinearRegression
LinearRegression()
0.9603182547438908
0.9184170849214232
array([[9281.30847068]])
array([27166.73682891])
Enter Years of Experience: 44
Estimated Salary for 44.0 years of experience is [[435544.30953887]]:

Result - Experiment 17

Linear regression was applied to the given dataset.